

Talking Open Data

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1 INTRODUCTION

Exploring Open Data remains an important challenge for the whole Open Data paradigm. Standard stock interfaces used by Open Data portals are anything but inspiring even for tech-savvy users, let alone those without an articulated interest in data science.

This demo showcases a cross-lingual Open Data search interface supporting natural language interactions via popular platforms like Facebook and Skype. Our data-aware chatbot answers search requests and suggests relevant open datasets.

3 USABILITY STUDY

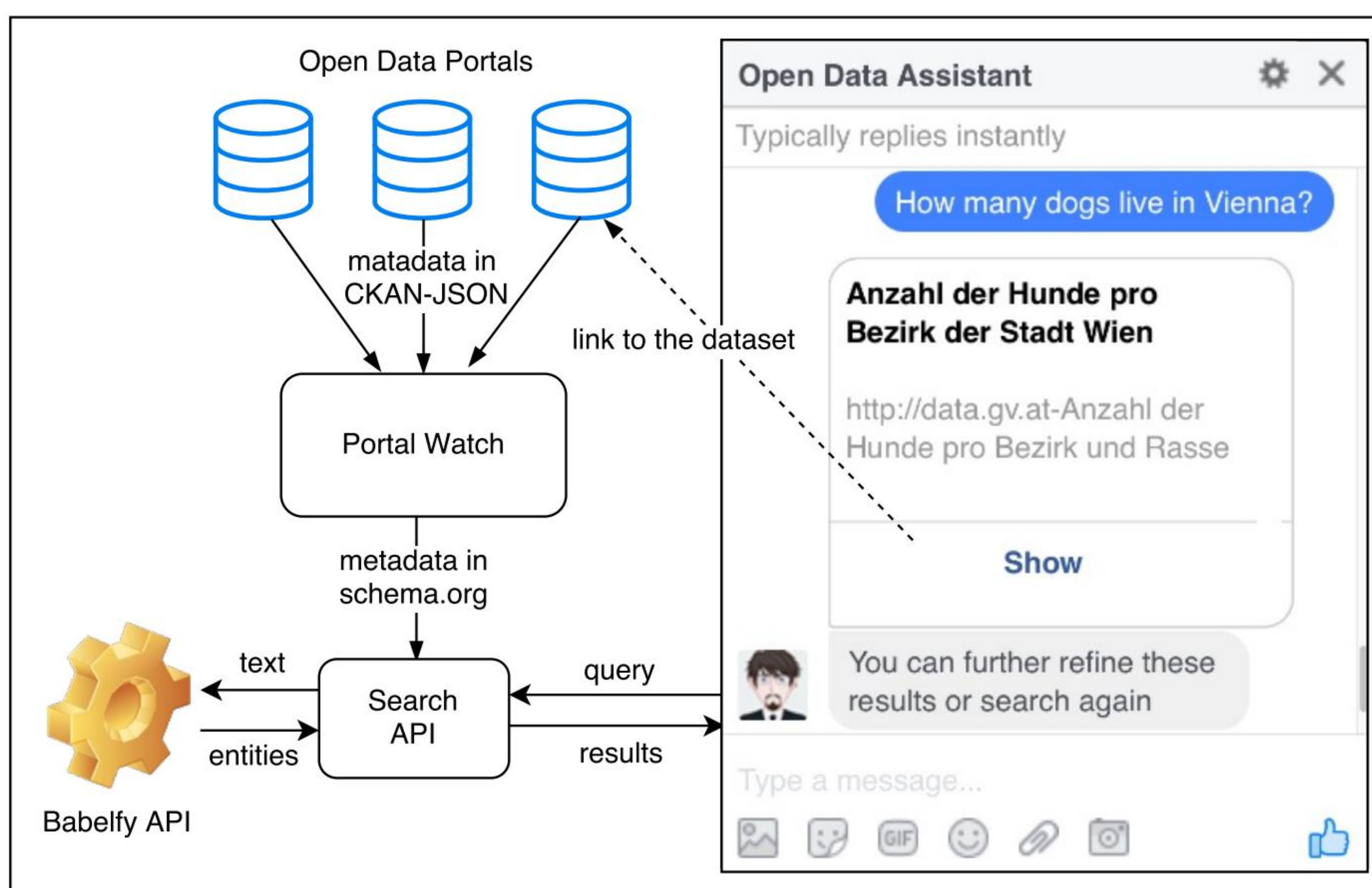
Tasks:

- Find climate statistics from different countries
- Reflect on experience of using the system

Results & suggestions:

- Useful, but in some cases limited in scope and functionality
- Complement with additional resources
- Adjust language of interaction
- Context-specific answers, e.g., geolocation-relevant queries

2 CHATBOT ARCHITECTURE



Approach:

1. Get portals' metadata in Schema.org from *Open Data Portal Watch*
2. Enrich metadata with concepts and entities extracted via the *Babelfly API*
3. Extract concepts from input queries
4. Refine search results by selecting co-occurring concepts and entities



- Integration of 7 different data portals and languages
- Search over metadata of 18k datasets

4 CONCLUSION & FUTURE WORK

- Extend set of indexed portals & datasets
- Search over datasets rather than metadata only
- Improve interaction and refinement features

PLATFORMS



<https://www.facebook.com/OpenDataAssistant/>



<https://join.skype.com/bot/6db830ca-b365-44c4-9f4d-d423f728e741>



REFERENCES

- Portal Watch* Neumaier, S., Umbrich, J., Polleres, A.: Automated Quality Assessment of Metadata across Open Data Portals. ACM J. Data Inform. Quality, 2016.
- Babelfy* Moro, A., Raganato, A., Navigli, R.: BabelNet: Entity Linking meets Word Sense Disambiguation: a Unified Approach. Trans. Ass. for Comp. Ling. (TACL), 2014.